



Estimation



Chapter Goals

After completing this chapter, you should be able to:

- Distinguish between a point estimate and a confidence interval estimate
- Construct and interpret a confidence interval estimate for a single population mean using both the Z and t distributions
- Form and interpret a confidence interval estimate for a single population proportion



Confidence Intervals

Content of this chapter

- Confidence Intervals for the **Population Mean, μ**
 - when Population Variance σ^2 is Known
 - when Population Variance σ^2 is Unknown
- Confidence Intervals for the **Population Proportion, $\hat{p}$** (large samples)



Definitions

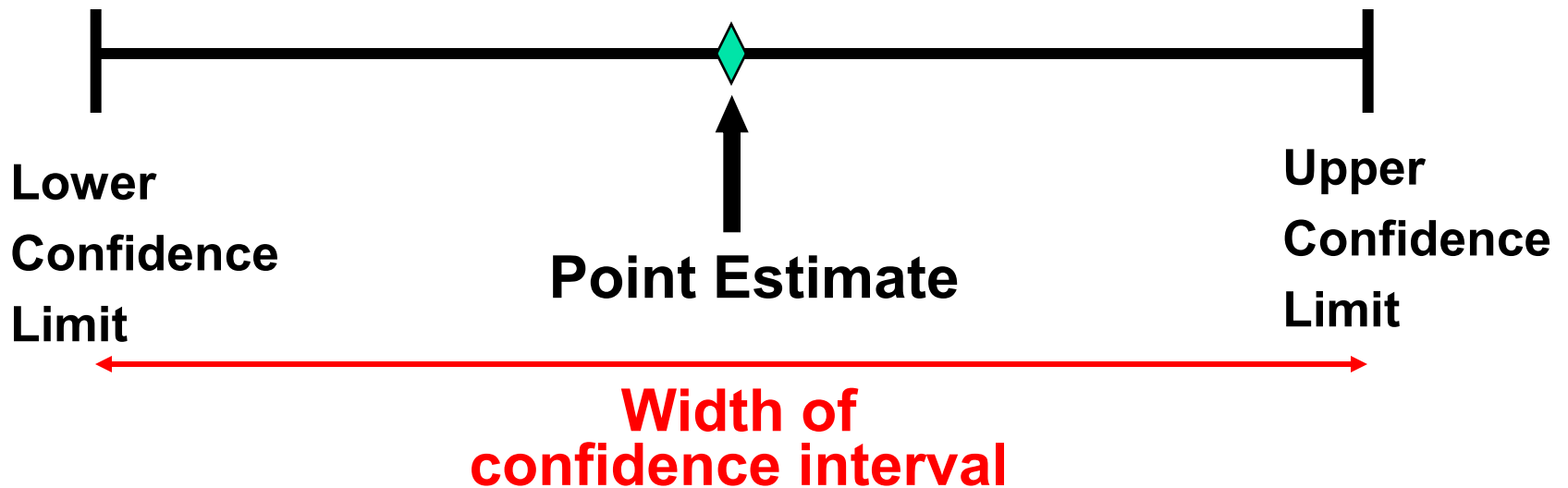
- An **estimator** of a population parameter is
 - a random variable that depends on sample information . . .
 - whose value provides an approximation to this unknown parameter

- A specific value of that random variable is called an **estimate**



Point and Interval Estimates

- A **point estimate** is a single number,
- a **confidence interval** provides additional information about variability





Point Estimates

We can estimate a Population Parameter ...		with a Sample Statistic (a Point Estimate)
Mean	μ	$\bar{x}$
Proportion	P	$\hat{p}$



Confidence Intervals

- How much uncertainty is associated with a point estimate of a population parameter?
- An **interval estimate** provides more information about a population characteristic than does a **point estimate**
- Such interval estimates are called **confidence intervals**



Confidence Interval Estimate

- An interval gives a **range** of values:
 - Takes into consideration variation in sample statistics from sample to sample
 - Based on observation from 1 sample
 - Gives information about closeness to unknown population parameters
 - Stated in terms of level of confidence
 - Can never be 100% confident

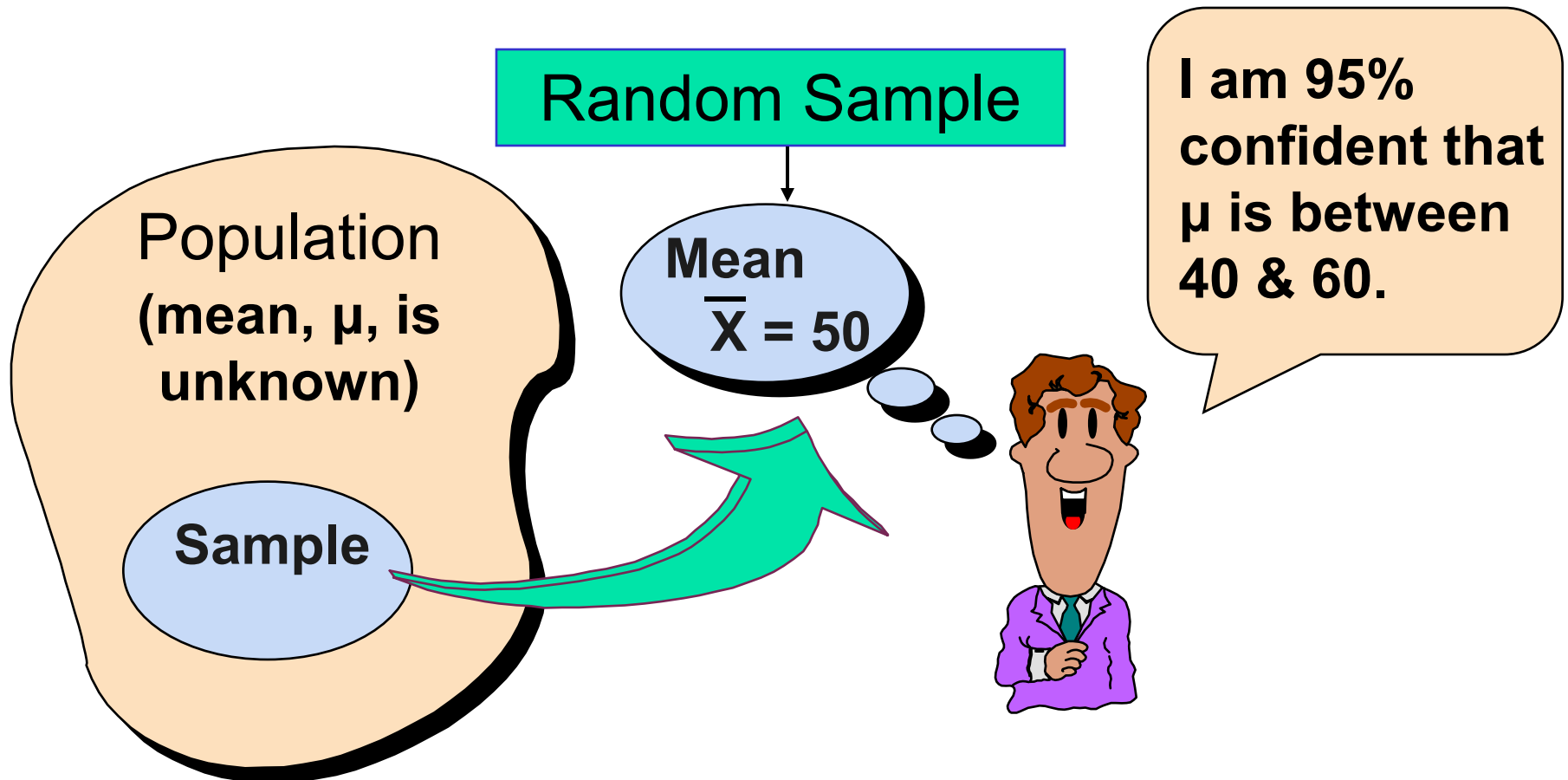


Confidence Interval and Confidence Level

- If $P(a < \theta < b) = 1 - \alpha$ then the interval from a to b is called a $100(1 - \alpha)\%$ confidence interval of θ .
- The quantity $(1 - \alpha)$ is called the confidence level of the interval (α between 0 and 1)
 - In repeated samples of the population, the true value of the parameter θ would be contained in $100(1 - \alpha)\%$ of intervals calculated this way.
 - The confidence interval calculated in this manner is written as $a < \theta < b$ with $100(1 - \alpha)\%$ confidence



Estimation Process





Confidence Level, $(1-\alpha)$

(continued)

- Suppose confidence level = 95%
- Also written $(1 - \alpha) = 0.95$
- A relative frequency interpretation:
 - From repeated samples, 95% of all the confidence intervals that can be constructed will contain the unknown true parameter
- A specific interval either will contain or will not contain the true parameter
 - No probability involved in a specific interval



General Formula

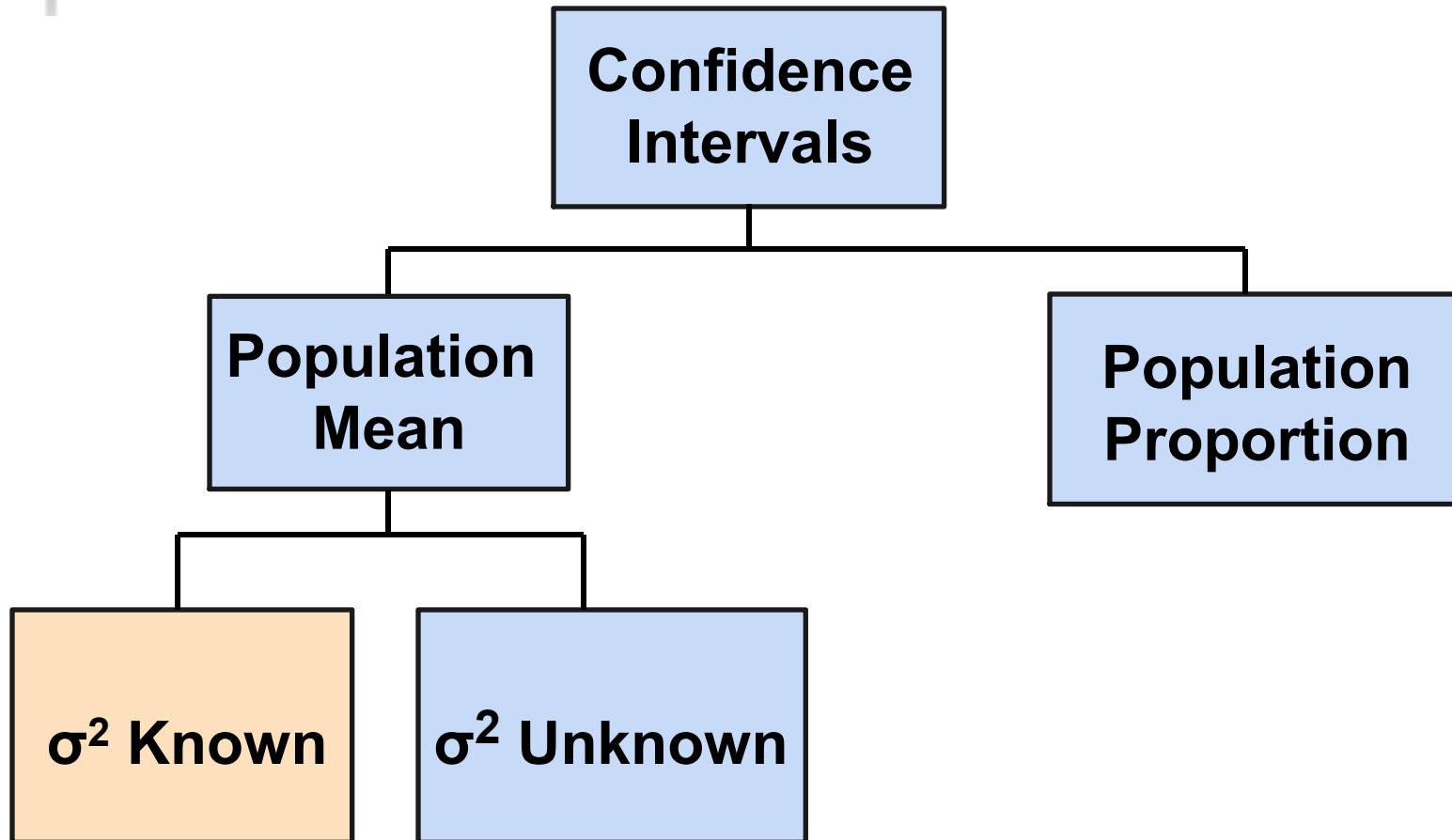
- The general formula for all confidence intervals is:

$$\text{Point Estimate} \pm (\text{Reliability Factor})(\text{Standard Error})$$

- The value of the reliability factor depends on the desired level of confidence



Confidence Intervals





Confidence Interval for μ (σ^2 Known)

- Assumptions
 - Population variance σ^2 is known
 - Population is normally distributed
 - If population is not normal, use large sample
- Confidence interval estimate:

$$\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

(where $z_{\alpha/2}$ is the normal distribution value for a probability of $\alpha/2$ in each tail)



Margin of Error

- The confidence interval,

$$\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} < \mu < \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

- Can also be written as $\bar{x} \pm \text{ME}$
where ME is called the **margin of error**

$$\text{ME} = z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

- The **interval width**, w , is equal to twice the margin of error



Reducing the Margin of Error

$$ME = z_{\alpha/2} \frac{\sigma}{\sqrt{n}}$$

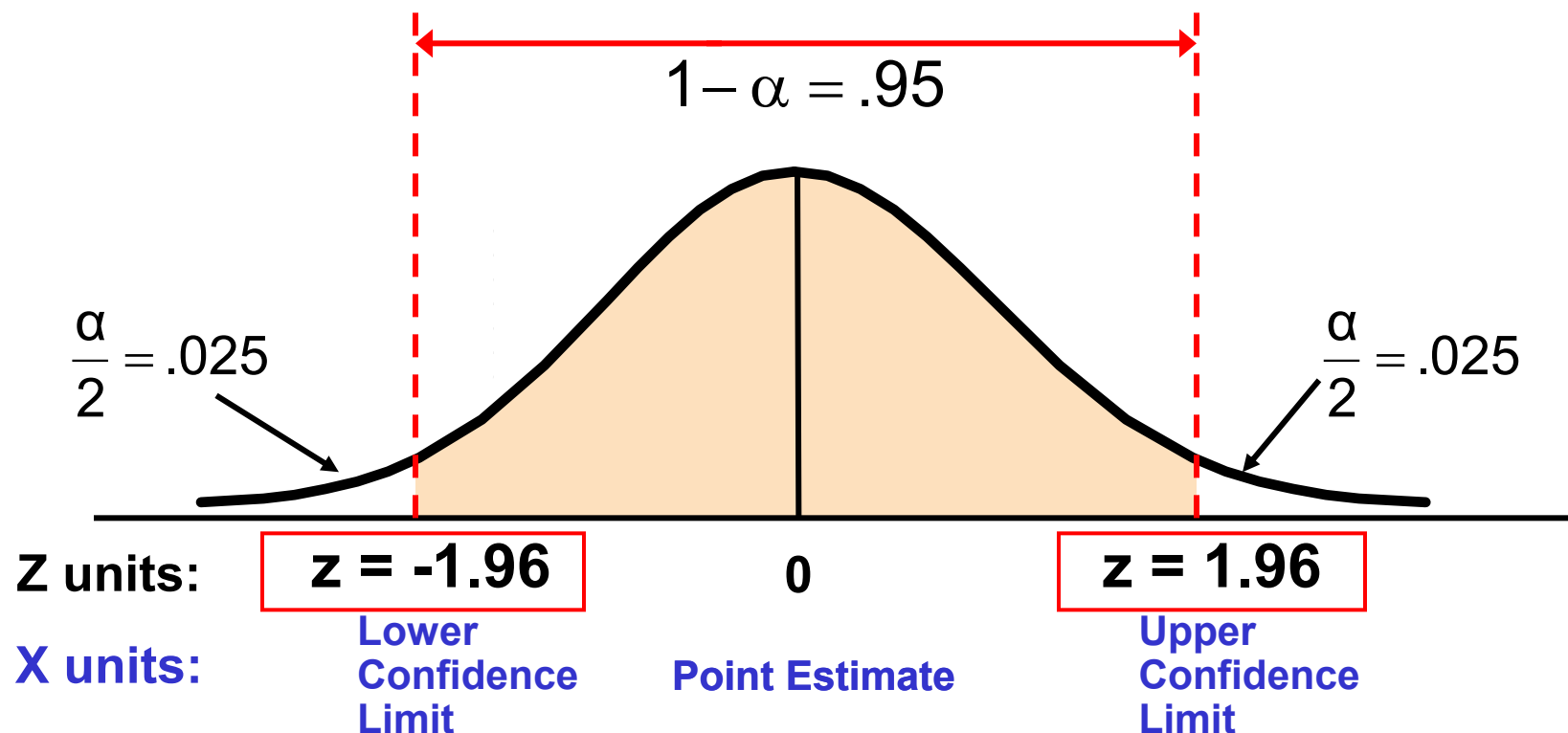
The margin of error can be reduced if

- the population standard deviation can be reduced ($\sigma \downarrow$)
- The sample size is increased ($n \uparrow$)
- The confidence level is decreased, $(1 - \alpha) \downarrow$



Finding the Reliability Factor, $z_{\alpha/2}$

- Consider a 95% confidence interval:



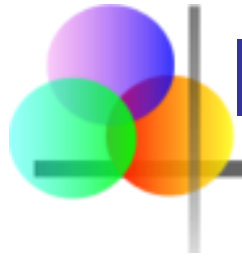
- Find $z_{.025} = 1.96$ from the standard normal distribution table



Common Levels of Confidence

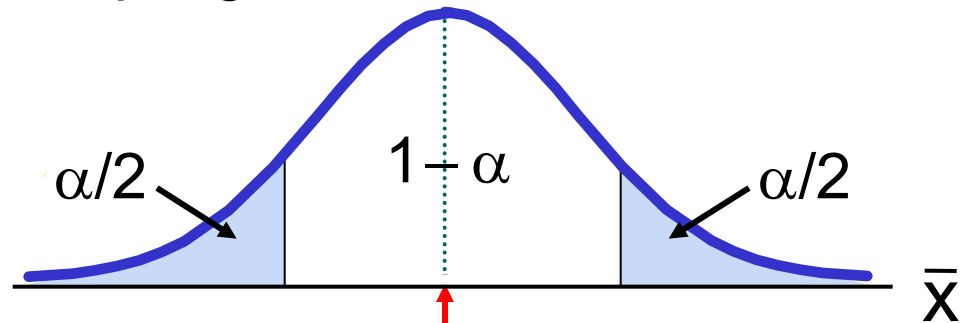
- Commonly used confidence levels are 90%, 95%, and 99%

Confidence Level	Confidence Coefficient, $1 - \alpha$	$Z_{\alpha/2}$ value
80%	.80	1.28
90%	.90	1.645
95%	.95	1.96
98%	.98	2.33
99%	.99	2.575
99.8%	.998	3.08
99.9%	.999	3.27



Intervals and Level of Confidence

Sampling Distribution of the Mean

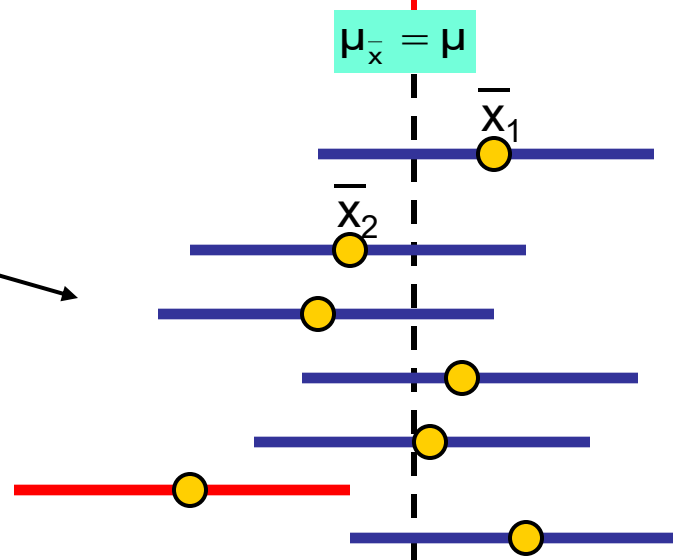


Intervals
extend from

$$\bar{x} - z \frac{\sigma}{\sqrt{n}}$$

to

$$\bar{x} + z \frac{\sigma}{\sqrt{n}}$$



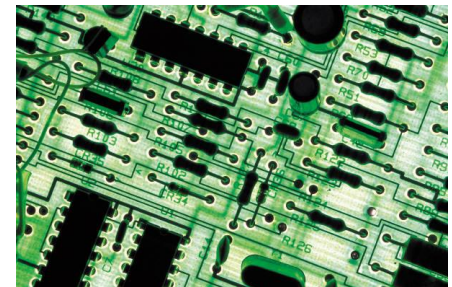
Confidence Intervals

100(1- α)%
of intervals
constructed
contain μ ;
100(α)% do
not.



Example

- A sample of 11 circuits from a large normal population has a mean resistance of 2.20 ohms. We know from past testing that the population standard deviation is 0.35 ohms.
- Determine a 95% confidence interval for the true mean resistance of the population.





Example

(continued)

- A sample of 11 circuits from a large normal population has a mean resistance of 2.20 ohms. We know from past testing that the population standard deviation is .35 ohms.

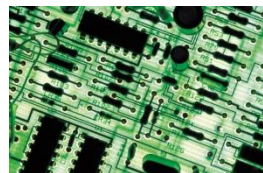
- **Solution:**

$$\bar{x} \pm z \frac{\sigma}{\sqrt{n}}$$

$$= 2.20 \pm 1.96 (.35/\sqrt{11})$$

$$= 2.20 \pm .2068$$

$$1.9932 < \mu < 2.4068$$





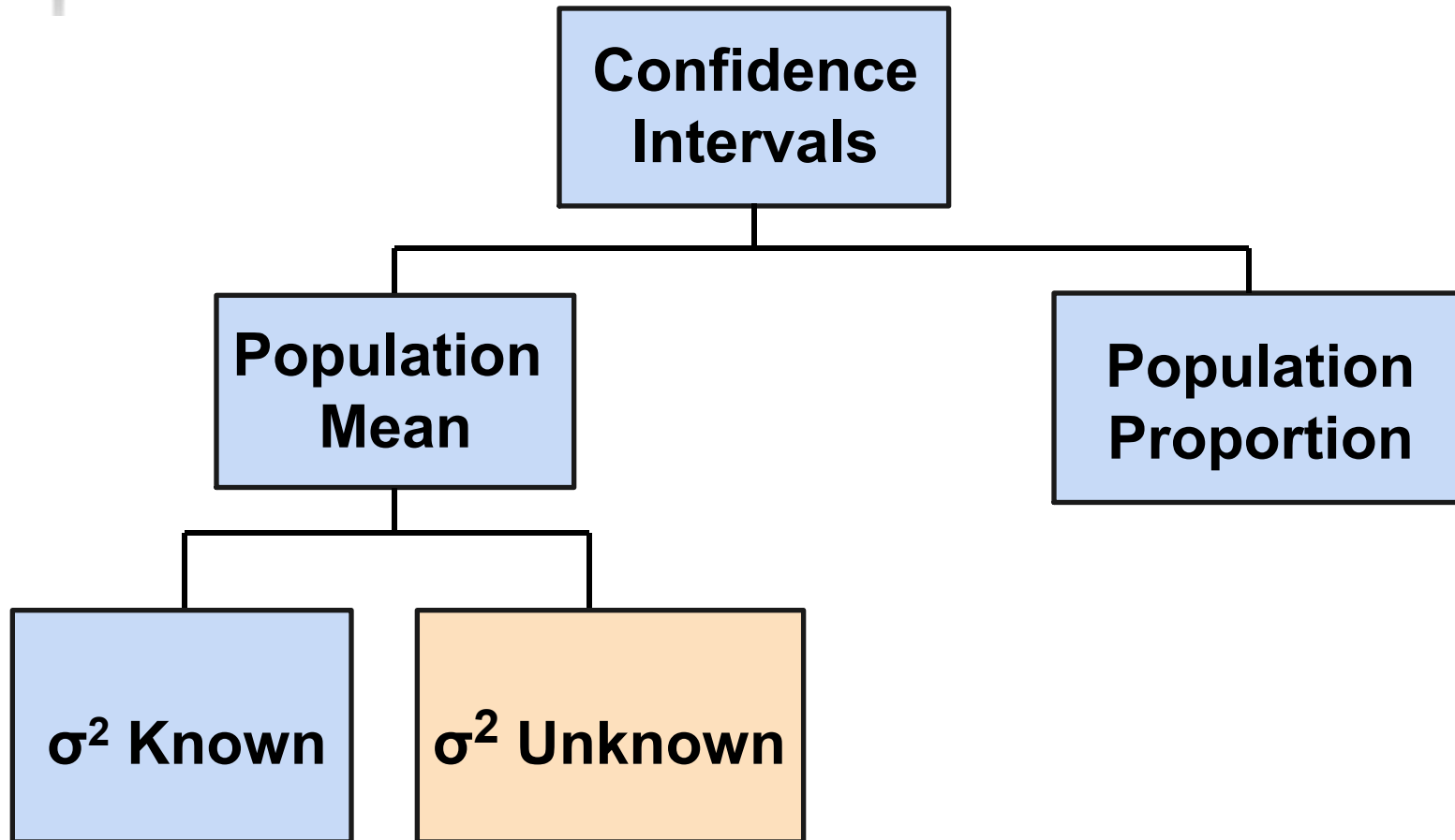
Interpretation

- We are 95% confident that the true mean resistance is between 1.9932 and 2.4068 ohms
- Although the true mean may or may not be in this interval, 95% of intervals formed in this manner will contain the true mean





Confidence Intervals





Student's t Distribution

- Consider a random sample of n observations
 - with mean $\bar{x}$ and standard deviation s
 - from a normally distributed population with mean μ
- Then the variable

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}}$$

follows the **Student's t distribution** with $(n - 1)$ degrees of freedom



Confidence Interval for μ (σ^2 Unknown)

- If the population standard deviation σ is unknown, we can substitute the sample standard deviation, s
- This introduces extra uncertainty, since s is variable from sample to sample
- So we use the t distribution instead of the normal distribution



Confidence Interval for μ (σ Unknown)

(continued)

- Assumptions
 - Population standard deviation is unknown
 - Population is normally distributed
 - If population is not normal, use large sample
- Use Student's t Distribution
- Confidence Interval Estimate:

$$\bar{x} - t_{n-1, \alpha/2} \frac{S}{\sqrt{n}} < \mu < \bar{x} + t_{n-1, \alpha/2} \frac{S}{\sqrt{n}}$$

where $t_{n-1, \alpha/2}$ is the critical value of the t distribution with $n-1$ d.f. and an area of $\alpha/2$ in each tail

$$P(t_{n-1} > t_{n-1, \alpha/2}) = \alpha/2$$



Student's t Distribution

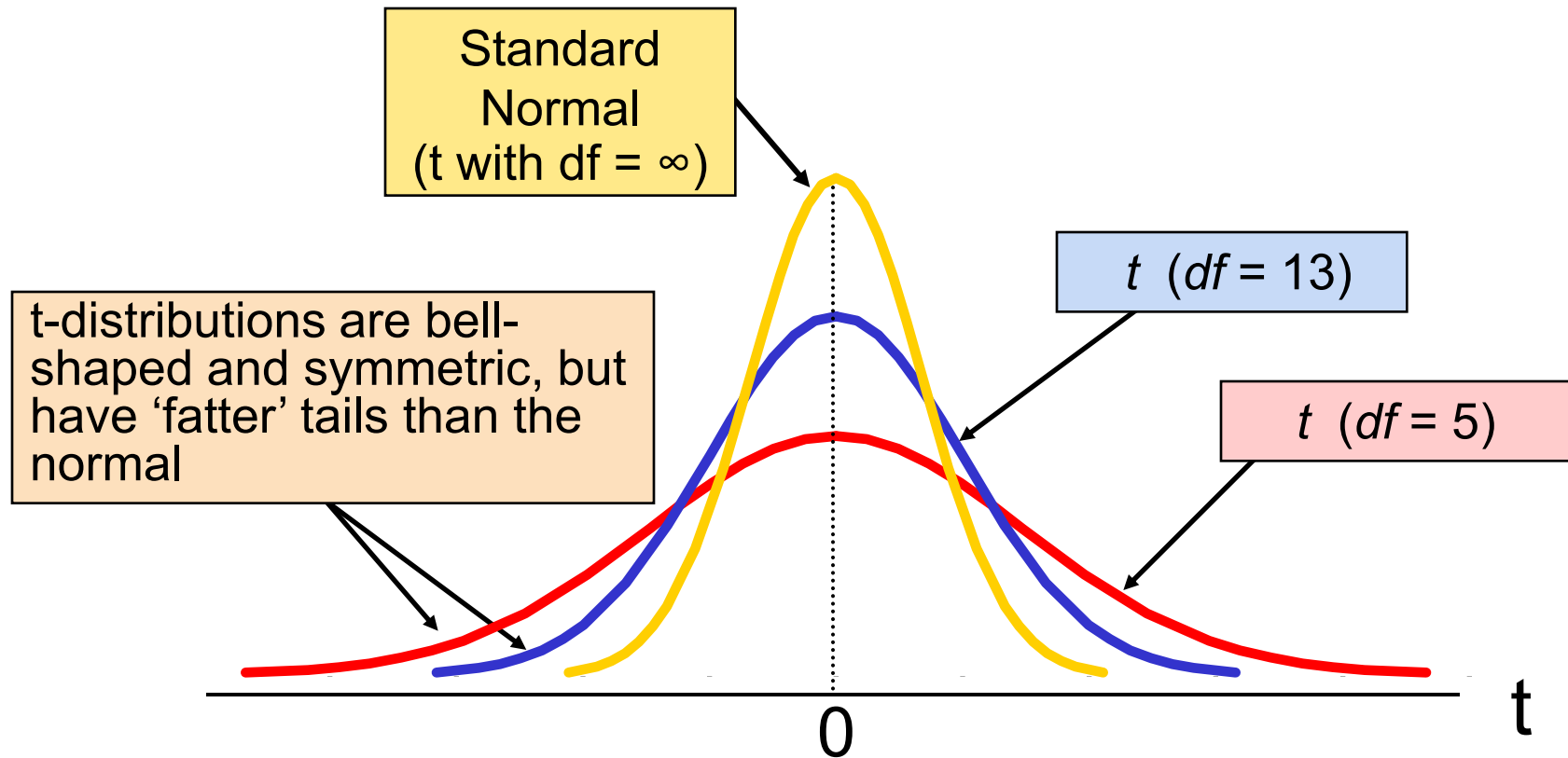
- The t is a family of distributions
- The t value depends on **degrees of freedom (d.f.)**
 - Number of observations that are free to vary after sample mean has been calculated

$$\text{d.f.} = n - 1$$



Student's t Distribution

Note: $t \rightarrow Z$ as n increases



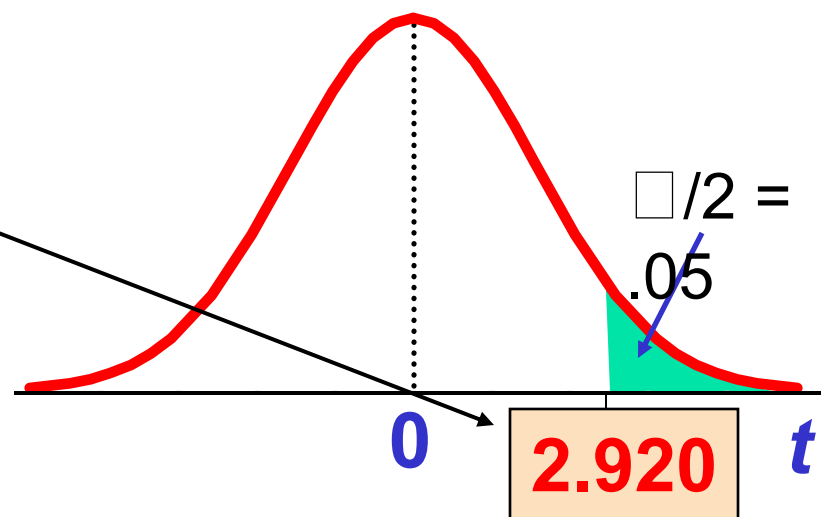


Student's t Table

Upper Tail Area			
df	.10	.05	.025
1	3.078	6.314	12.706
2	1.886	2.920	4.303
3	1.638	2.353	3.182

Let: $n = 3$
 $df = n - 1 = 2$
 $\square = .10$
 $\square / 2 = .05$

The body of the table contains t values, not probabilities





t distribution values

With comparison to the Z value

Confidence Level	t (10 d.f.)	t (20 d.f.)	t (30 d.f.)	Z
.80	1.372	1.325	1.310	1.282
.90	1.812	1.725	1.697	1.645
.95	2.228	2.086	2.042	1.960
.99	3.169	2.845	2.750	2.576

Note: $t \rightarrow Z$ as n increases



Example

A random sample of $n = 25$ has $\bar{x} = 50$ and $s = 8$. Form a 95% confidence interval for μ

- d.f. = $n - 1 = 24$, so $t_{n-1, \alpha/2} = t_{24, .025} = 2.0639$

The confidence interval is

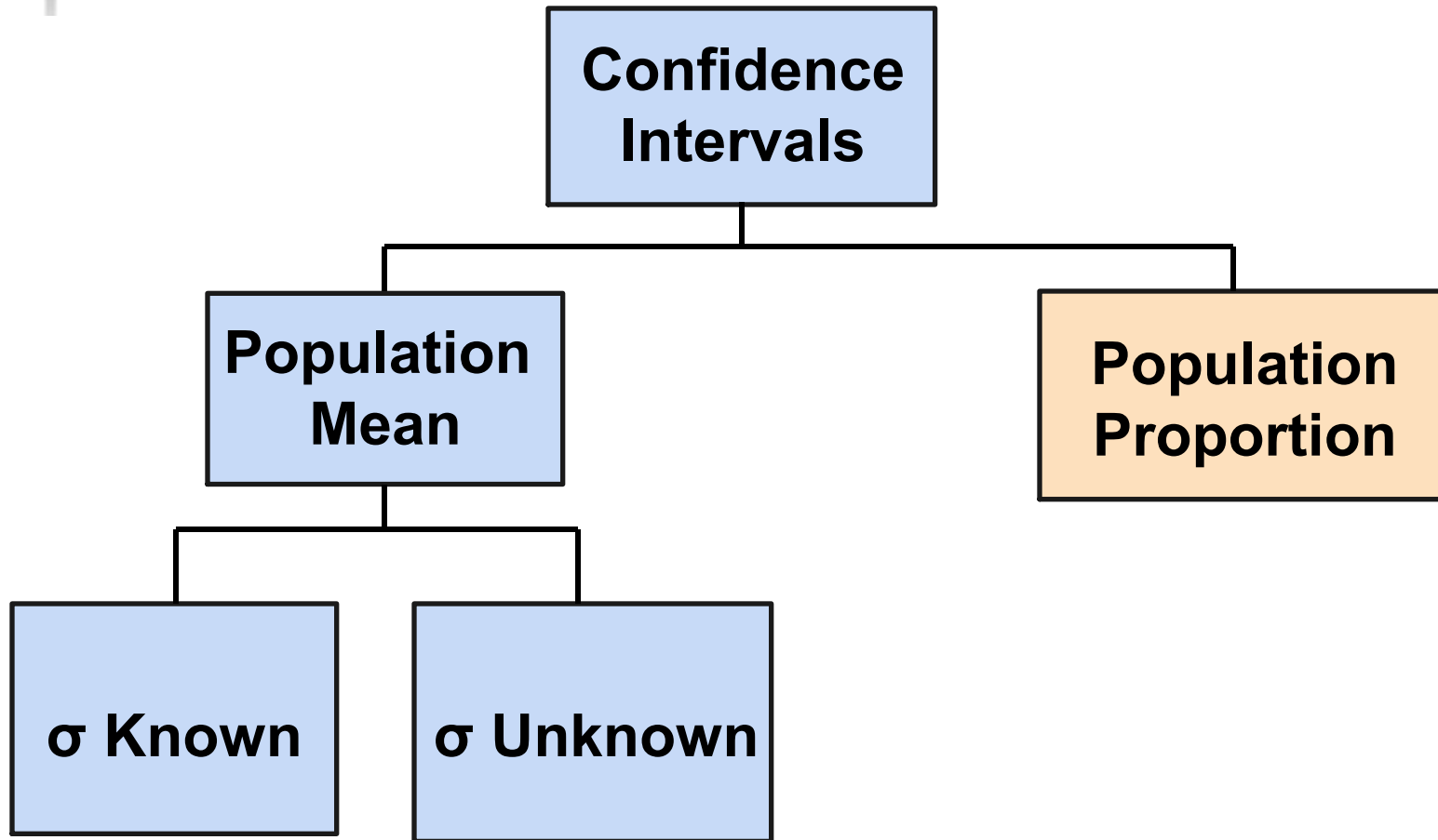
$$\bar{x} - t_{n-1, \alpha/2} \frac{s}{\sqrt{n}} < \mu < \bar{x} + t_{n-1, \alpha/2} \frac{s}{\sqrt{n}}$$

$$50 - (2.0639) \frac{8}{\sqrt{25}} < \mu < 50 + (2.0639) \frac{8}{\sqrt{25}}$$

$$46.698 < \mu < 53.302$$



Confidence Intervals





Confidence Intervals for the Population Proportion, p

- An interval estimate for the population proportion (P) can be calculated by adding an allowance for uncertainty to the sample proportion ($\hat{p}$)



Confidence Intervals for the Population Proportion, p

(continued)

- Recall that the distribution of the sample proportion is approximately normal if the sample size is large, with standard deviation

$$\sigma_P = \sqrt{\frac{P(1-P)}{n}}$$

- We will estimate this with sample data:

$$\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$



Confidence Interval Endpoints

- Upper and lower confidence limits for the population proportion are calculated with the formula

$$\hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} < P < \hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

- where
 - $z_{\alpha/2}$ is the standard normal value for the level of confidence desired
 - $\hat{p}$ is the sample proportion
 - n is the sample size



Example

- A random sample of 100 people shows that 25 are left-handed.
- Form a 95% confidence interval for the true proportion of left-handers





Example

(continued)

- A random sample of 100 people shows that 25 are left-handed. Form a 95% confidence interval for the true proportion of left-handers.

$$\hat{p} - z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} < P < \hat{p} + z_{\alpha/2} \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\frac{25}{100} - 1.96 \sqrt{\frac{.25(.75)}{100}} < P < \frac{25}{100} + 1.96 \sqrt{\frac{.25(.75)}{100}}$$

$$0.1651 < P < 0.3349$$





Interpretation

- We are 95% confident that the true percentage of left-handers in the population is between
16.51% and 33.49%.
- Although the interval from 0.1651 to 0.3349 may or may not contain the true proportion, 95% of intervals formed from samples of size 100 in this manner will contain the true proportion.





Chapter Summary

- Introduced the concept of confidence intervals
- Discussed point estimates
- Developed confidence interval estimates
- Created confidence interval estimates for the mean (σ^2 known)
- Introduced the Student's t distribution
- Determined confidence interval estimates for the mean (σ^2 unknown)
- Created confidence interval estimates for the proportion